Smart Soil Testing and Decision Support System for Cashew Using Nitrogen Analysis

Project ID – 25-26J-175

Project Proposal Report

Mithushh P- IT22180902

Bachelor of Science (Hons) Degree in Information Technology
Specializing in

Information Technology

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Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously publish or written by another person expect where the acknowledgement is made in the text.

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The supervisor/s should certify the proposal report with the following declaration.
The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

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Signature of the Co-Supervisor Date

Ms. Suranjini Silva

ABSTRACT

Farmers need healthy soil to grow good crops, but many do not know the real condition of their soil. In the real world, manual soil testing is slow. Costly, and requires laboratory facilities that are not available in rural areas. This is a serious problem in the cashew cultivation because there is no proper soil testing, farmers may use unwanted fertilizers, or waste money. Bad soil management can lower the cashew production and affect the quality of the harvest. Many beginner level farmers cannot afford to test their soil in the lab many times, so they have to make decisions based on guesswork or what they did before. So, farmers can make better decisions and grow healthier cashew trees.

So, the purpose of the research is a smart soil health analysis system with a decision support system using soil level testing. The idea is to make soil testing quick and easy with the help of artificial intelligence and image processing. Farmers can take a clear photo of the soil with a mobile phone. The system will then study the photo to check important details such as soil type, color, texture, moisture and nitrogen levels, these are the key factors that affect how well crops grow. A trained machine learning model will process this information and give results immediately, so farmers do not have to wait for days or weeks like in manual testing. In addition to showing soil health, the system also shows the decision support for the farmers. Farmers will guide them by recommending which fertilizers to use, how much to use, and which cashew plants are most suitable for their soil. By doing this, the system helps farmers reduce costs, and improve their harvest

After testing, the system gives clear advice. It tells farmers which cashew variety is best, what fertilizers to use, and how to improve the soil. If the soil is not suitable, it suggests simple fixes for example, adding lime for acidic soil, adding compost for sandy or alkaline soil, or adding sand or gypsum for clay soil. These solutions are easy for farmers to follow. Farmers can test their soil again after making changes. The system will then give updated results and new advice. This way, farmers can always check and improve their soil at a low cost and grow cashew trees more successfully.

Keywords

Soil Testing, Artificial Intelligence (AI), Image Processing, Nitrogen Level Analysis, Decision Support System

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LIST OF ABBREVIATIONS

Key Words	Meaning
AI	Artificial Intelligence
DL	Deep Learning
ML	Machine Learning
pH	Potential of Hydrogen

Table 1 List of Abbreviations

INTRODUCTION

1.1 Background

Cashew is an important crop that provides income for many farmers around the world. Cashew nuts are popular for food, snacks, and oils, and the demand for them is growing every year [1]. In countries like Sri Lanka, India, Vietnam, and several African nations, cashew farming helps thousands of families earn money and improve their living standards [2]. Because of this, keeping cashew farms healthy and productive is very important.

One of the biggest problems for cashew farmers is testing their soil quickly and correctly. If the soil is not healthy, cashew trees may grow slowly, produce fewer nuts, and get sick easily [1]. Traditional soil testing methods can help, but they are often slow, expensive, and require laboratory facilities that many farmers in rural areas cannot reach [2]. Because of this, many farmers grow cashews without knowing the real condition of their soil. This leads to low production, poor-quality nuts, and less profit.

Traditional soil testing methods can help, but they are often slow, expensive, and require laboratory facilities that many farmers in rural areas cannot reach [5]. In some cases, farmers must travel long distances to submit soil samples, wait for weeks to get results, and pay high fees for laboratory tests. Because of this, many farmers grow cashews without knowing the real condition of their soil. This leads to low production, poor-quality nuts, and less profit.

To solve this problem, new technologies like artificial intelligence and machine learning can be used. Farmers can take a photo of their soil using a smartphone or a low-cost camera. The system then examines the soil's color, texture, and moisture to determine its type and overall health [3]. A special feature of the system is that it can also estimate the nitrogen level in the soil. Nitrogen is very important for plant growth, and it is closely related to soil color and organic matter. By using this connection, the system can tell if nitrogen is low, medium, or high, giving farmers quick and useful information about soil fertility [4].

This smart soil analysis system is not only for testing but also works as a decision-making tool. It suggests which cashew variety is most suitable for the farmer's soil. If the soil is not good enough, the system gives easy and practical advice to improve it. For example, it may suggest adding lime for acidic soil, compost for sandy or alkaline soil, or sand or gypsum for clay soil [5]. After making these improvements, farmers can test their soil again by uploading new pictures, and the system will give updated results and advice. This helps farmers gradually improve their soil quality step by step [6].

The system will be available as both a web and mobile application, so farmers can use it on smartphones, tablets, or computers. Through the app, they can upload photos of their soil, get instant results about soil type and nitrogen level, and receive recommendations about the best cashew variety to plant. Over time, farmers can save their results, track changes in the soil, and compare old and new data. This makes it easier to make better farming decisions [7].

In the future, the system can be made stronger by adding Internet of Things (IoT) technology. For example, wireless sensors can help with irrigation and soil moisture monitoring [8], water management systems can control water use [9], and rainfall prediction models using machine learning can help farmers plan their activities according to weather [10]. These improvements support climate-smart agriculture, which protects the environment while giving good crop yields [11]. By using this smart soil analysis system, farmers can save money, reduce the need for costly lab tests, grow healthier cashew trees, and produce more nuts. In the end, the system will increase farmers' income, make their work easier, and help the growth of the agricultural economy [12].

1.2 Literature survey

Many researchers have studied how technology can make soil testing easier, faster, and cheaper for farmers. Different methods have been tried, such as using smartphones, satellites, and deep learning models. These studies show good results, but each method also has some limitations that need to be improved in future research.

One study developed a low-cost soil testing system that uses only a smartphone and paper test strips [12]. Farmers place a small amount of soil on the strip, and then the phone camera takes a photo. A machine learning model checks the color on the strip to understand if the soil pH is acidic, neutral, or basic. In field experiments, the app gave correct results about 92% of the time, which is very close to lab results. The system is simple, cheap, and does not need any advanced tools, so farmers can test soil directly in the field instead of sending samples to laboratories. This saves both time and money and makes soil testing possible even in rural areas where labs are not available. It also helps farmers take quick decisions about which crops to grow or which fertilizers to use.

Another study used satellite images and AI models to test soil texture, such as sandy loam [13], loam, and clay loam. The researchers trained two machine learning models: Decision Tree and Support Vector Regression. When they compared results, the SVR model worked much better. It was able to predict sandy loam soil with 100% accuracy, while for other textures the accuracy was between 81% and 90%. This shows that satellite data combined with AI can give very reliable soil information. The big advantage is that this method can test large areas of land at once, instead of just small soil samples. Farmers and

governments can use this to make soil maps and guide farming decisions over wide regions, saving a lot of time and cost compared to traditional testing.[13]

A study tested a deep learning model that used soil photos to check sand, clay, and silt levels. The system combined a CNN (Convolutional Neural Network) with Random Forest. It reached about 99% accuracy and gave very close results to real lab data (R^2 above 0.98) [14]. This shows that photos from a normal phone or camera can be enough to test soil texture.

Another research study applied a deep learning approach using a Convolutional Neural Network (CNN) to directly classify soil into different types based solely on images. The CNN architecture consisted of 8 layers, which allowed the model to automatically learn and extract important features from the soil images, such as texture, color, and pattern. During the training process, the model analyzed a large dataset of soil pictures, learning to distinguish subtle differences between soil types. After training, CNN achieved an impressive accuracy of approximately 95% in correctly classifying the soil types. This high level of performance demonstrates that deep learning can be a powerful and efficient method for soil testing, enabling accurate soil type identification using only images without the need for traditional chemical tests or laboratory equipment.

A study focused on estimating soil nitrogen (Total Nitrogen, TN) using satellite reflectance data combined with additional texture features extracted from soil images. The researchers tested two machine learning models: Random Forest (RF) and XGBoost. Both models analyzed image data and satellite measurements to predict the nitrogen content in the soil. The results showed that the XGBoost model performed better, achieving a high accuracy with a coefficient of determination (R^2) of 0.84, while the Random Forest model reached a lower R^2 of 0.72. These results indicate that AI and machine learning methods can be effective tools for predicting soil nitrogen levels, offering farmers a faster and cheaper alternative to traditional laboratory testing. By using satellite and image-based data, farmers can monitor soil health more efficiently and make better decisions for fertilization and crop management.

Researchers have used different technologies to test soil quickly and cheaply. Smartphones with paper strips and machine learning can check soil pH. Satellites with AI models like Decision Tree and SVR can study soil texture. Cameras or photos with deep learning models such as CNN and Random Forest can find sand, clay, and silt. AI models like Random Forest and XGBoost can predict soil nitrogen. For my project, I will use MobileNetV2/V3 with soil images to make fast and easy soil testing for farmers.

COMPARISON TO OLD MODELS

Old Model	New Model
Previous research used traditional lab tests or earlier AI models to analyze soil. These methods required sending soil samples to a laboratory or using complex algorithms that were slow and not practical for farmers in the field. They could measure pH, texture, or nitrogen, but usually took a long time, were costly, and did not provide instant or crop-specific recommendations for cashew farming.	The proposed system uses MobileNetV2 and image processing to analyze soil directly from photos taken with a smartphone or low-cost camera. It can instantly identify soil type, pH, moisture, and classify nitrogen levels as low, medium, or high. The system also gives cashew-specific advice, including the best variety to plant and fertilizer recommendations. This approach is fast, low-cost, and practical for field use, helping farmers make better decisions, improve soil health, and increase cashew yield.

Table 2 compared to old model

1.1Research Gap

Many studies have explored soil testing using AI and machine learning. Some research used smartphones and paper strips to measure soil pH, giving results close to lab tests. Other studies applied satellite images and AI models to identify soil texture and type, such as sandy loam, loam, and clay loam, achieving high accuracy. Deep learning methods like Convolutional Neural Networks (CNN) have also been used to classify soil types and measure sand, clay, and silt levels from images, showing that photos alone can provide reliable soil information. Similarly, some research predicted soil nitrogen using satellite data and machine learning models like Random Forest and XGBoost, with XGBoost performing better. However, most of these studies either require lab tests, satellite data, or heavy computation, which can be costly and less practical for farmers in the field. There is limited work on using mobile-friendly deep learning models, such as MobileNetV2 and MobileNetV3, to predict soil nitrogen levels directly from images. Specifically, classifying nitrogen into low, medium, and high levels using a smartphone has not been fully explored. This creates an opportunity to develop a fast, low-cost, and easy-to-use mobile system for farmers, allowing them to monitor soil nitrogen directly in their fields without expensive equipment or waiting for lab results.

1.2 Research Problem

The main problem in this study is that cashew farmers do not have a quick, easy, and low-cost way to check their soil health. Traditional soil testing requires sending samples to a laboratory, which takes a long time, costs money, and is not practical for regular testing. Many farmers cannot access these labs, and as a result, they often plant cashew trees in soil that is not suitable. This leads to poor tree growth, low fruit production, and higher chances of disease.

Currently, there is no simple, cashew-specific soil testing system that farmers can use directly in the field to get instant results and actionable recommendations. Existing systems are either designed for other crops or focus only on identifying soil type, without providing specific guidance for cashew farming. They also often require expensive equipment or technical skills, which many small-scale farmers do not possess.

To address this, the proposed system will use MobileNetV2 or MobileNetV3, lightweight and efficient deep learning models, to analyze soil photos captured with a smartphone or low-cost camera. These models can classify soil type, estimate nutrient levels such as nitrogen, and provide accurate, real-time results suitable for field use.

2 OBJECTIVES

2.1 Main Objectives

The main goal of my project component is to analyze soil and identify its nitrogen level using AI and image processing. Farmers can take a photo of their soil with a smartphone or low-cost camera, and the system will determine the soil type and classify nitrogen levels as low, medium, or high. This helps farmers understand their soil's nutrient status quickly without sending samples to a laboratory and make better decisions about fertilization and crop management.

The system also includes three other parts. Pest control helps identify and prevent pest problems in cashew trees. Disease analysis detects early signs of diseases to protect the trees and reduce losses. Cashew nut grading evaluates the quality of harvested nuts, helping farmers sell them at the best price. Together, these four components form a completely smart farming solution that supports healthier trees, higher yields, and increased income for farmers.

2.2 Specific Objectives

1. Develop a user-friendly web application for instant soil health testing.

2. To use AI and image processing to analyze soil color, texture, and moisture from photos.

3. To Train deep learning models to detect soil type and estimate pH range accurately.

4. To provide cashew-specific recommendations for crop variety, fertilizer use, and soil improvement.

5. To test the system with real soil samples from different cashew-growing areas.

6. To collect farmers feedback to improve system accuracy and usability.

7.To implement MobileNetV2 or MobileNetV3 as the core deep learning model for lightweight and fast analysis.

8. To include nitrogen level estimation from soil images as a novel feature for better fertility assessment.

3 METHODOLOGY

3.1 System overview Diagram

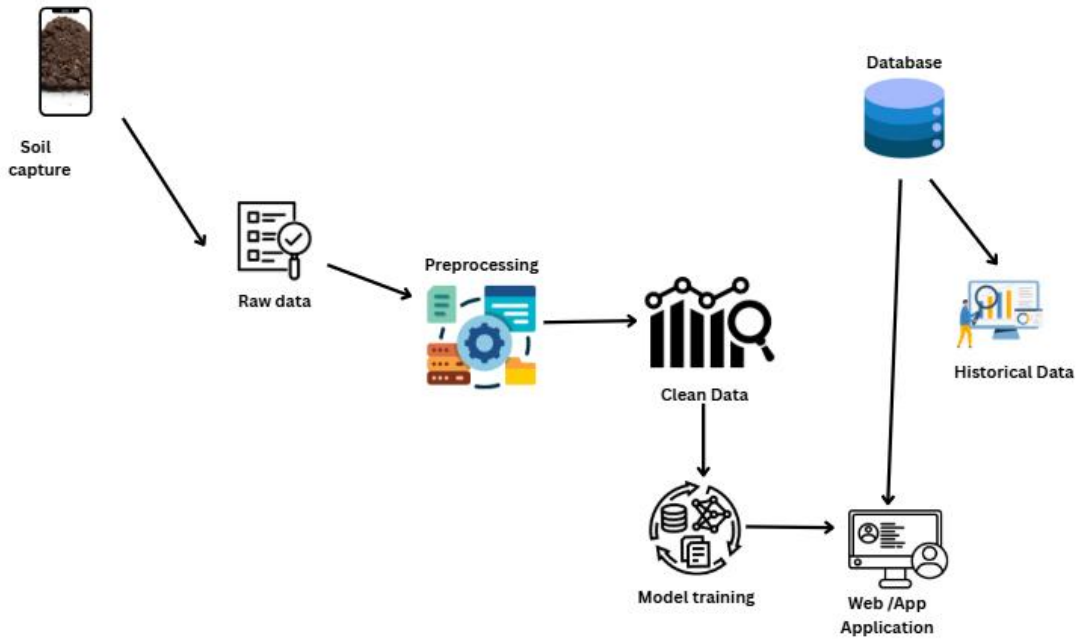

Figure:3.1 Complete System Architecture Diagram

Soil samples will be collected from different cashew-growing areas. For each sample, photos will be taken using smartphones or low-cost cameras under different lighting conditions. At the same time, soil properties such as pH, moisture, and type will be measured using portable soil testing kits. These results will be used as ground truth labels for training and testing the AI models.

The soil images will be prepared before training. This includes resizing images, adjusting brightness and contrast, removing background noise, and labeling them with the correct soil properties. Data augmentation methods such as rotation, flipping, and zooming will be applied to create more training samples and improve model accuracy.

Lightweight deep learning models such as MobileNetV2 and MobileNetV3 will be trained to analyze soil images. The models will learn to detect properties such as soil color, texture, moisture, and nitrogen levels. Different models of architecture will be compared, and the most accurate and efficient ones will be selected. The performance of the model will be validated using both the collected dataset and independent test samples.

A web-based application will be developed to allow farmers to upload soil photos and receive instant results. The system will display soil properties (type, nitrogen level) along with cashew-specific recommendations such as the best variety to plant, suitable fertilizers, and soil improvement advice. The system will also store soil test history so that farmers can track changes in their soil over time.

The system will be tested with real farmers and their soil samples in selected cashew-growing regions. The results produced by the AI system will be compared with laboratory test results to check accuracy. Farmers' feedback on ease of use and usefulness of the recommendations will also be collected. This feedback will help improve the system before its official release.

3.2 Gantt chart

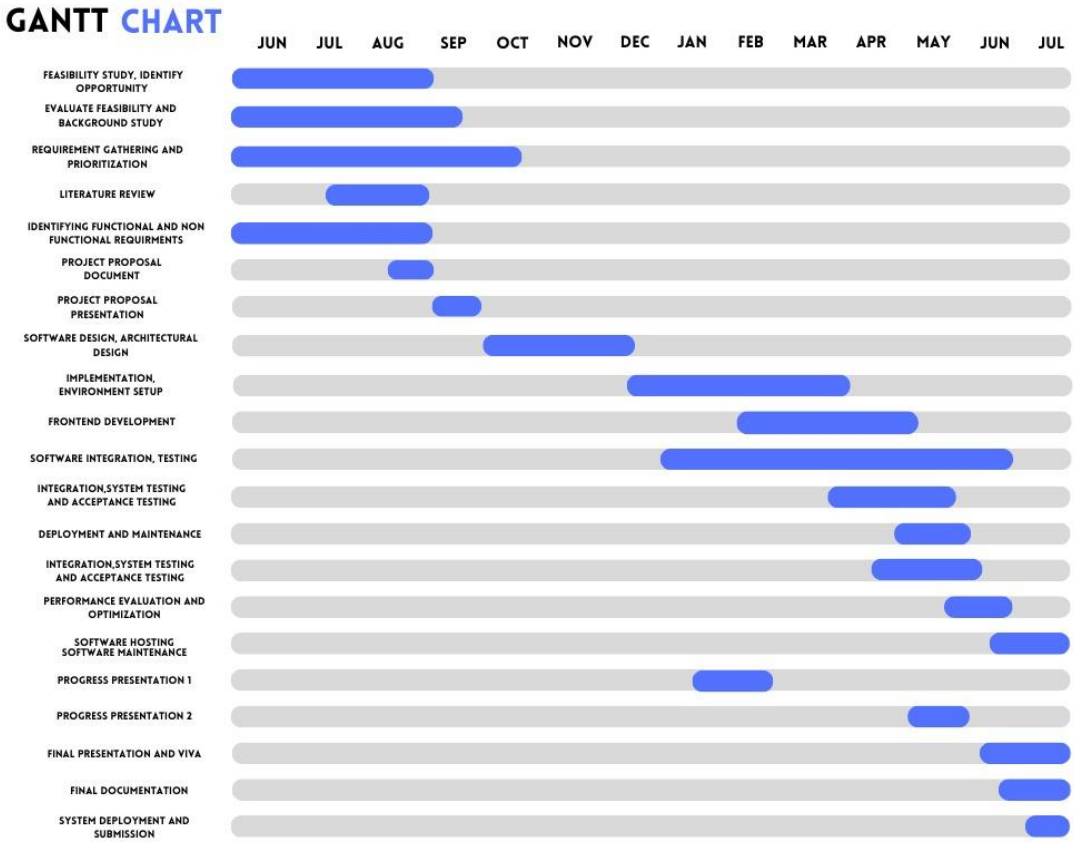

Figure: 4.1 Gantt chart

The Gantt chart Figure 4.1 shows the timeline for the whole project. For my part, I will mainly work on frontend development, software integration and testing, and system testing and acceptance. Frontend development is scheduled from January to March, where I will build the user interface for the soil testing app. Integration and testing happen from February to May, where I will connect the frontend with the backend and check if everything works correctly. The final system deployment and submission are in May, which means the project will be finished and ready to use by then. This chart helps to plan my tasks step by step and finish the project on time.

3.3 Work breakdown structure

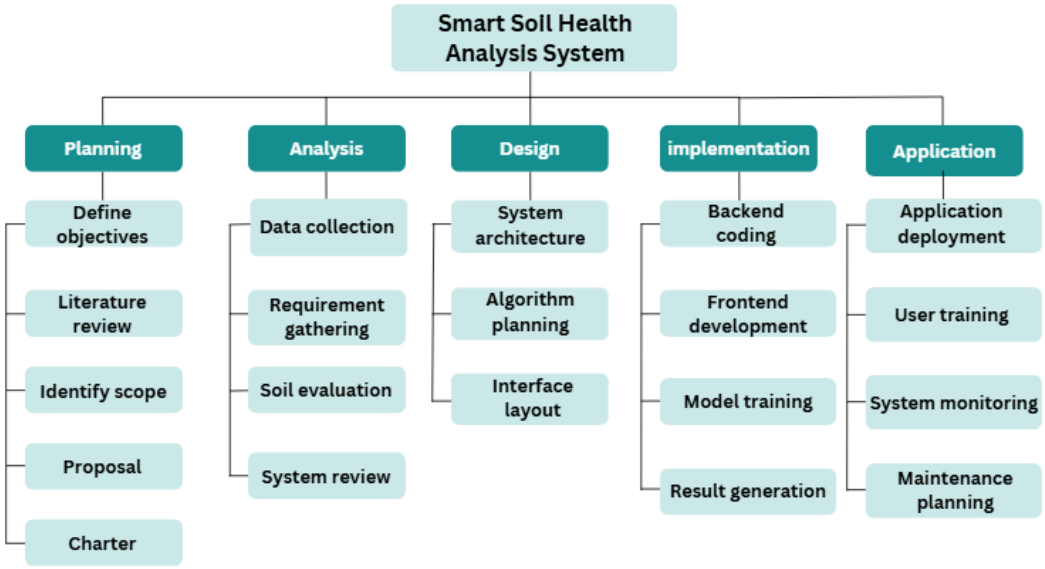

Figure: 4.2 Work breakdown structure

It is a step-by-step process that starts with planning, then moves to collecting and analyzing data, followed by designing and building the system, and ends with deploying and maintaining it in Figure 4.2 Work Breakdown Structure.

4 Project requirements

4.1 Functional Requirements

- Farmers can enter soil information manually or upload a soil photo.
- The system can check soil type and nitrogen level.
- System Features can tell what kind of soil it is (like sandy, clay, or red soil) and nitrogen level (like Low, Medium, High).
- System Features recommends the best cashew variety for the soil.
- System Features suggest fertilizers, watering, and how to improve the soil.
- Proposed System Functions gives advice for planning and caring for cashew trees.
- Proposed System Functions creates a simple report with results and suggestions.
- Provides an approximate nitrogen level based on soil color and texture
- The system uses MobileNetV2 or MobileNetV3 to analyze soil images accurately

4.2 Nonfunctional Requirements

1. Performance: The system should give results quickly (within seconds or minutes).
2. Usability: The system should be easy to use, even for farmers with little technical knowledge.
3. Reliability: The system should work correctly all the time without crashing.
4. Accessibility – It should be available on mobile phones and computers, anytime in the field.
5. Scalability – The system should handle many users and soil tests at the same time.
6. Accuracy – Soil analysis results should be correct and reliable.

4.3 Technology Selection

1. Programming language
   - Python: A programming language used to create machine learning models, process images, and handle the system's backend operations.

2. Machine Learning and Deep Learning
   - TensorFlow: A popular free tool used to create and train deep learning models, like CNNs, to check the quality and defects in cashew nuts.
   - MobileNetV2 / MobileNetV3

3. Frontend
   - HTML/CSS/JS
   - React JS

4. Database
   - SQLite or MySQL

5. Desing Tools
   - UI design: Figma
   - Diagram Design: Draw.io, Canva

6. Version Control
   - Git/GitHub

7. Deployment and Hosting
   - AWS/Google Cloud/Azure

7 CONCLUSION

The Smart Soil Health Analysis System is a useful tool for cashew farmers to check their soil quickly, cheaply, and accurately. Using AI and image processing, the system can identify soil type, pH, moisture, and overall health, giving farmers all the information they need to make better decisions. It also provides practical advice on which cashew variety to plant, what fertilizers to use, and how to improve soil quality.

The system works with smartphones or low-cost cameras, making it accessible to small-scale farmers in rural areas who may not have easy access to laboratories. By using this system, farmers can grow healthier cashew trees, increase their yield, save time and money, and reduce waste of resources like water and fertilizers. It encourages sustainable farming practices by helping farmers maintain and improve soil fertility over time.

Moreover, the system can be expanded in the future to include other smart farming tools, such as automated irrigation, crop monitoring, or fertilizer recommendation systems. This integration will make farming easier, more precise, and more productive. Overall, this project shows how modern technology like AI and data-driven analysis can support farmers, improve agricultural outcomes, and increase profits, while also promoting sustainable and environmentally friendly farming practices.

8 BUDGET AND BUDGET JUSTIFICATION

Item	Description	Cost Estimation
Deploy in Play store	Deploy the app in the play store	32,000 LKR
Google Cloud	12-months	9100 LKR
Google Colab	For model training	15,100 LKR
Total		56,200 LKR

Table 3 Budget and Budget Justification

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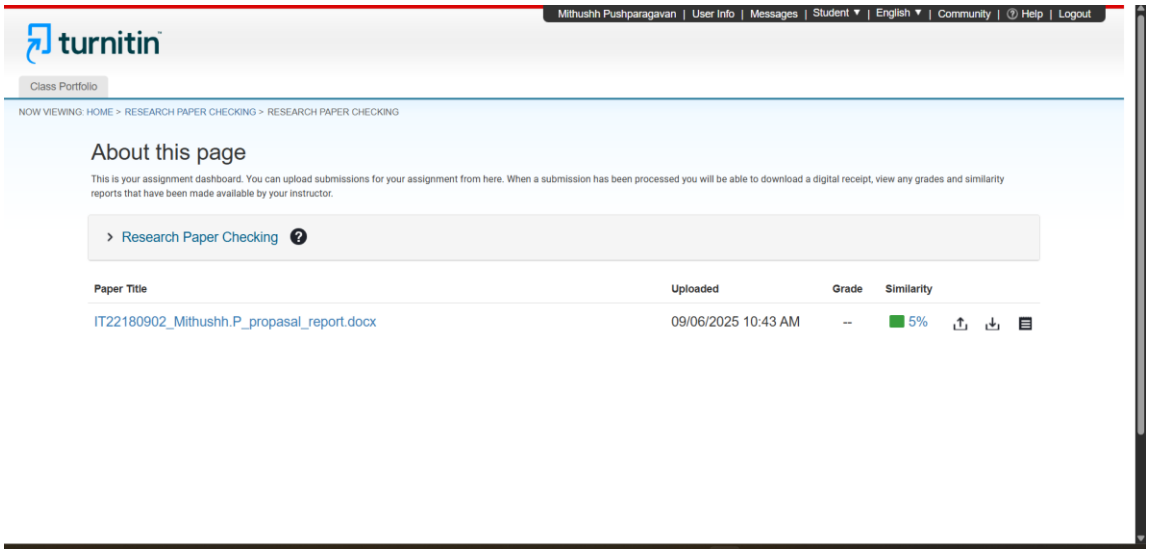
10 Appendices



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